the field lines around and around, causing them to fling plasma upward and downward in much the same manner as the third method, to form two jets. The jets shoot out along the hole's spin axis and their direction thus is firmly anchored to the hole's gyroscopic spin. This method was conceived of by Blandford soon after he received his Ph.D. in Cambridge, together with a Cambridge graduate student, Roman Znajek, and it thus is called the *Blandford–Znajek process.*

The Blandford–Znajek process is especially interesting, because the power that goes into the jets comes from the hole's enormous rotational energy. (This should be obvious since it is the hole's spin that causes space to swirl, and the swirl of space that causes the magnetic field lines to rotate, and the field lines' rotation that flings plasma outward.)

How is it possible, in this Blandford–Znajek process, for the hole's horizon to be threaded by magnetic field lines? Such field lines would be a form of "hair" that can be converted into electromagnetic radiation and be radiated away, and therefore, according to Price's theorem (Chapter 7), they *must* be radiated away. In fact, Price's theorem is correct only if the black hole is sitting alone, far from all other objects. The hole we are discussing, however, is not alone; it is surrounded by an accretion disk. If the field lines of Figure 9.7d pop off the hole, the lines going out the hole's northern hemisphere and those going out its southern hemisphere will turn out to be continuations of each other, and the only way these lines can then escape is by pushing their way out through the accretion disk's hot gas. But the hot gas will not let the field lines through; it confines them firmly into the region of space inside the disk's inner face, and since most of that region is occupied by the hole, most of the confined field lines thread through the hole.

Where do these magnetic field lines come from? From the disk itself. All gas in the Universe is magnetized, at least a little bit, and the disk's gas is no exception.[4] As, bit by bit, the disk's gas accretes into the hole, it carries its magnetic field lines with it. Upon nearing the hole, each bit of gas slides down its magnetic field lines and through the horizon, leaving the field lines behind, sticking out of the horizon and threading it in the manner of Figure 9.7d. These threading field lines, firmly confined by the surrounding disk, should then extract the hole's rotational energy by the Blandford–Znajek process.